

2025 AMC 10B Problem 5 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 5



Problem:

In $\triangle ABC$, $AB = 10$, $AC = 18$, and $\angle B = 130^\circ$. Let O be the center of the circle containing points A , B , C . What is the degree measure of $\angle CAO$?

Answer Choices:

- A. 20
- B. 30
- C. 40
- D. 50
- E. 60



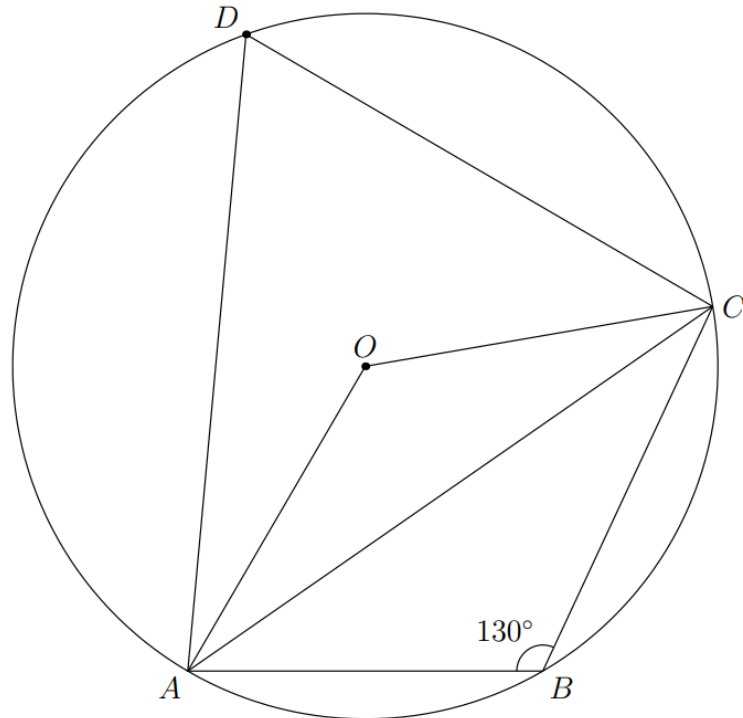
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Solution:

Let O be the circumcenter of $\triangle ABC$. Since $\angle ABC = 130^\circ$, we can say that $\triangle ABC$ is obtuse and hence contained within one half of the circle as shown.



Let D be a point on the opposite side of the arc $\widehat{ABC}$. Now we have a cyclic quadrilateral $ABCD$. As opposite angles in a cyclic quadrilateral add to 180° , we have $\angle ADC = 180^\circ - \angle ABC = 180^\circ - 130^\circ = 50^\circ$. Therefore, as central angles are twice the inscribed angles, we have

$$\angle AOC = 2\angle ADC = 2 \cdot 50^\circ = 100^\circ$$

Finally, we see that the $\triangle AOC$ is isosceles and therefore

$$\angle CAO = \angle ACO = \frac{180^\circ - 100^\circ}{2} = \boxed{\text{(C) } 40}$$

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